

Method for Performing Multilingual OCR and Transcribing Textual Documents

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1 Purpose

The goal of this methodology is to process images (e.g. binarization, fix orientation, content selection, DPI settings, etc.) and then perform multilingual object character recognition (OCR) capable of recognizing multiple languages in order to output machine-readable PDF files that can be translated into other languages.

2 System Requirements

Please note that these instructions are intended for a Windows operating system. The same procedure can be followed for MacOS, but the file locations and instructions intended specifically for File Explorer will be different.

Applications

ScanTailor

3 Methodology

*Note: **Bold** typeface indicates project-specific values. These fields can vary depending on the scan settings and the text itself.*

3.1 SCANTAILOR

System Requirements:

- *ScanTailor*

1. *If this is your first time using ScanTailor, please follow the ScanTailor Install instructions in Section 4.1.*
2. Open a new project in ScanTailor.
3. Fix the DPI of all pages to **400 x 400**.
 - a. In the “Fix DPI” window (“Tools” → “Fix DPI...”), select the “All Pages” tab.
 - b. Click “All Pages” in the smaller window. Once you have made your selection, the options at the bottom of the window should be available for you to change.
 - c. In the dropdown, select “**400 x 400**”.
 - d. Click apply.
4. Remove any duplicate pages from the project.
5. Fix orientation of all pages such that they are in a portrait layout.

6. Set “Split Pages” to “Split Manually” on “All Pages” so that the software doesn’t automatically split the pages in the next step.
7. Run “Deskew” on the “Automatic” setting on “All Pages”. Fix manually as needed such that text lines are parallel to on-screen gridlines.
8. Use “Select Content” on “All Pages” to highlight all text, images, page numbers, and annotations on each page. Manually adjust as needed.
9. Use the “Margins” tab to add the margins to each page, omitting the scanner background.
 - a. Decouple the “Top” and “Bottom,” “Left” and “Right” fields.
 - b. Adjust the margins in each field until the entire page is within the selected region.
10. In the “Output” tab, do the following:
 - a. Apply an “Output Resolution” of **400dpi** to “All Pages”.
 - b. Apply “Black and White” to “All Pages”.
 - c. Leave all other settings at their default.
 - d. Run.
 - e. *Note: ScanTailor keeps the file names, but loses the metadata information. The metadata will need to be re-appended to the exported files.*
11. “File” → “Save Project As...” to save the project file and output in an appropriate folder (usually the input directory).
12. The output files are located in the input directory in a new folder labeled “OUT”.

3.2 GOOGLE GEMINI

System Requirements:

- Internet access and access to Google Gemini

Optical Character Recognition

1. Open a new chat in Google Gemini.
2. Upload one ScanTailor TIFF output file and copy the following prompt:
 - a. “Run a multilingual (**French, Vietnamese, and Chinese**) OCR on this document and output the text exactly as written.”
 - b. The **bold** text above should be changed depending on the languages included in your document.
3. Copy and paste the output into a text file (e.g. Word Document, Google Doc, .txt file).

- a. Make sure that the text is readable, keeping the italics and other special font formats.
 - b. Correct dashes as necessary (these used to be words that broke over the line).
4. Continue this process for all of the text pages in the chapter.

English Transcription

1. Open a new chat in Google Gemini.
2. Upload one ScanTailor TIFF output file and copy the following prompt:
 - a. “Translate the **French** text below to **English**, but leave the **Vietnamese and Chinese characters the same**. The translation should retain the original text format, including italics, bold font, and paragraph separations such that it can be copy and pasted.”
 - b. The **bold** text above should be changed depending on the languages included in your document.
3. Copy and paste the output into a text file (e.g. Word Document, Google Doc, .txt file).
 - a. Make sure that the text is readable, keeping the italics and other special font formats.
 - b. Correct dashes as necessary (these used to be words that broke over the line).
4. Continue this process for all of the text pages in the chapter.

4 Installation Instructions

4.1 SCANTAILOR INSTALL

“ScanTailor is an interactive post-processing tool for scanned pages – page splitting, deskewing, borders, and more” (ScanTailor, n.d.).

1. Navigate to [ScanTailor](#).
2. Select the blue “Downloads & binaries” button. You should now be on the Downloads page. Navigate to the “Github project page” using the link on the website.
3. On GitHub, scroll down to the README file. Below the “Installation and Tips” header, select the “installer” link.
4. Below the “Installation and first start” header, follow the necessary instructions to install and start the application.

Bibliography

ScanTailor. (n.d.). *About ScanTailor* [ScanTailor]. Retrieved July 6, 2026, from <https://scantailor.org/about/>